Rational periodic loci of nonconstant quadratic Hénon maps over function fields

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Abstract

We classify every rational periodic point of the quadratic Hénon map $(x, y) \mapsto (y, y^2 + c - ax)$ over $k(t)$, for any field k of characteristic different from two, any nonzero constant a, and any nonconstant polynomial c. The existence of a periodic point forces $c = -P^2 + C$, with $C \in k$ unique and $P \in k[t]$ unique up to sign. Every periodic coordinate then equals P or $-P$ plus a constant. An injectively labelled, two-sheeted lift of an eight-state partial permutation turns this restriction into an exact rationality criterion. Direct case analysis gives seven cycle types, including characteristic-three branches of least periods five and eight. The total number of rational periodic points is at most fourteen, with equality precisely in characteristic three at $a = -1$ and $c = -P^2 - 1$: one four-cycle and two five-cycles coexist. Outside characteristics two and three the sharp bound is eight when -1 is a square in k, and six otherwise. Every period greater than two requires $a^4 = 1$. The classification counts ordinary points over $k(t)$, without passage to its completion or a sign quotient.

1 Introduction

An exact rational periodic locus asks more than whether periodic points are finite or their periods are bounded. For a polynomial automorphism over a function field, a local symbolic description can allow many itineraries that do not come from rational functions. We determine this rationality restriction for the entire family

$$H_{a,c}(x, y) = (y, y^2 + c - ax), \qquad a \in k^*, \quad c \in k[t] \setminus k, \quad \operatorname{char} k \neq 2. \tag{1.1}$$

The domain throughout is $k(t)^2$. In particular, the nonconstant parameter is polynomial, while the determinant is constant.

The reduction is elementary but global. At a finite polynomial prime, a coordinate with maximal pole order contradicts the quadratic recurrence. At infinity, all coordinates in all periodic orbits have the same positive degree. Factoring the difference of their squares then puts every coordinate in $\{P + b, -P + b : b \in k\}$ for one common nonconstant polynomial P. The recurrence determines the offsets from neighboring signs. Only sixteen signed labels remain, and their exact transition graph decides which are recurrent.

The main conclusions are the following.

1. The parameter must have the unique centered form $c = -P^2 + C$, where uniqueness of P is understood up to sign. A seven-row atlas then constructs every rational cycle, including all overlaps.

2. The sharp universal total is fourteen rational periodic points. The unique equality locus is $\operatorname{char} k = 3$, $a = -1$, $C = -1$; the ordinary periods there are four and five.

3. Outside characteristics two and three, the sharp bound is eight or six according as -1 is or is not a square in k. Longer-than-two periods in any allowed characteristic require $a^4 = 1$.

No hypothesis of perfection, algebraic closure, finite cardinality, ordering or local compactness is imposed on k.

Arithmetic bounds and conventions. Canonical heights and non-Archimedean escape are established tools for Hénon arithmetic. Ingram's function-field finiteness and bad-place bounds are prior results [2]; neither the existence of such bounds nor the local escape method is a contribution here. His normalization is $\varphi_\alpha(x, y) = (\alpha y, x + f(y))$. For $L(x, y) = (-ax, y)$ one has

$$LH_{a,c}L^{-1}(x, y) = (-ay, x + y^2 + c).$$

Thus the source's special value $\alpha = 1$ corresponds to our $a = -1$, not to the conservative value $a = 1$. Our comparison uses the introduction and relevant local arguments of arXiv:1111.3609v1, especially its stated quadratic routing; the journal metadata in the bibliography does not assert that its final proof text was read. Those accessed statements give finiteness and bounds, not the present all-determinant equality locus and every signed cycle.

A local horseshoe is not the global rational locus. Allen, DeMark and Petsche study $\varphi_{A,B}(X, Y) = (A + BY - X^2, X)$ over complete, locally compact non-Archimedean fields of odd residue characteristic [1]. Their horseshoe theorem gives a full two-symbol bilateral shift in the appropriate region $|A| > \max(1, |B|^2)$ with A a square. The change $J(x, y) = (-y, -x)$ gives $JH_{a,c}J^{-1} = \varphi_{-c,-a}$. For finite odd k the completion $k((1/t))$ fits their field setting, but a shift over the completion does not select the itineraries in $k(t)$. For arbitrary k the completion need not be locally compact. Our graph is a partial permutation precisely because it imposes this global rationality restriction. The coding method itself is classical.

Nearby repository results. The earlier unpublished repository note C412 on conservative integral quadratic Hénon maps already uses sign/constant-offset equations at $a = 1$. We credit that encoding, as well as its real separation and finite small-parameter completion; C412 is not represented as a journal publication. The present proof replaces those integer-size arguments by across-cycle polynomial-degree rigidity and resolves every allowed constant determinant and characteristic. Constant-coefficient finite-field height distributions concern a different parameter family and observable. These distinctions identify the claim studied here; they are not a claim of an exhaustive priority search.

Section 2 states the atlas and reconstruction. Sections 3–5 prove the global reduction, exact graph and all-field exhaustion. Section 6 retains every row collision to prove the sharp bounds.

2 The exact cycle atlas

Fix the data in (1.1). A cycle means an ordinary orbit of least positive period. We do not identify points by a sign involution and do not count fixed-scheme multiplicities. Write

$$K_a = \frac{(a + 1)^2}{4}.$$

Binary words below are words over the set $\{0, 1\}$. Bits are mapped to k only when they occur in a field equation.

Theorem 2.1 (Rationality and reconstruction)**.** *If $H_{a,c}$ has a periodic point in $k(t)^2$, then*

$$c = -P^2 + C, \qquad P \in k[t] \setminus k, \quad C \in k. \tag{2.1}$$

When such a representation exists, C is unique and P is unique up to sign. Fix either choice of P. Every rational cycle is given by Table 1 and the reconstruction rule

$$\sigma_{i+1} = \sigma_i(-1)^{\delta_i}, \qquad y_i = \sigma_i P + \frac{(-1)^{\delta_i} + a(-1)^{\delta_{i-1}}}{2}, \qquad (y_i, y_{i+1}) \in k[t]^2. \tag{2.2}$$

Here (δ_i) repeats the row's transition word in both directions, and σ_0 is either sign. A row contributes if and only if its conditions and its value of C hold in k. Simultaneously applicable rows contribute disjoint point sets and are added. If (2.1) does not exist, or if it exists but no row applies, there are no rational periodic points.

Word	Additional condition	C	Cycles	Least period
0	none	K_a	2	1
1	none	$-3K_a$	1	2
01	$a^2 = 1$	$K_a - 1$	1	4
001	$a^2 = -1$	K_a	1	6
011	$a = 1$	-1	2	3
00011	$\operatorname{char} k = 3$, $a = -1$	-1	2	5
0111	$\operatorname{char} k = 3$, $a = -1$	1	1	8

Table 1: Every rational cycle under the standing assumptions $a \neq 0$, $c \notin k$ and $\operatorname{char} k \neq 2$. Use the common centered parameter (2.1); add all applicable rows. An even number of ones in the word gives two cycles of the word length. An odd number gives one cycle of twice that length. These are ordinary point periods.

Theorem 2.2 (Sharp totals and determinant restriction)**.** *There are at most fourteen rational periodic points. Equality holds exactly when*

$$\operatorname{char} k = 3, \qquad a = -1, \qquad c = -P^2 - 1 \quad \text{for some } P \in k[t] \setminus k. \tag{2.3}$$

These maps have one four-cycle and two five-cycles. If $\operatorname{char} k \notin \{2, 3\}$, the sharp bound is eight when -1 is a square in k, and six otherwise. Eight points occur exactly at $a^2 = -1$ and $C = K_a$, as two fixed points and a six-cycle. In the nonsquare case, six points are attained at $a = 1$ and $C = -1$. In every allowed characteristic, a point of period greater than two requires $a^4 = 1$.

In particular, odd degree of c precludes every rational periodic point. The converse of this last observation alone is false: even a centered square parameter can miss every row of the atlas. All sufficiency assertions in the theorem refer to the explicit points in (2.2), without extending k.

3 Global polynomial and two-branch rigidity

Write an ordinary periodic orbit as (y_i, y_{i+1}), with indices cyclically extended. Its recurrence is

$$y_i^2 + c = y_{i+1} + a y_{i-1}. \tag{3.1}$$

The inverse map is $((x^2 + c - y)/a, x)$, so this recurrence is valid without excluding periods one and two or inserting a preperiodic tail. Polynomial degree always means degree in t, with $\deg 0 = -\infty$ in degree inequalities.

Lemma 3.1 (Finite-pole escape)**.** *Every rational periodic coordinate lies in* $k[t]$*.*

Proof. For an irreducible polynomial $\pi \in k[t]$, let $v_\pi(\pi) = 1$. If some coordinate has a pole at π, choose one of maximal pole order $M > 0$ within its finite orbit. At that index the left side of (3.1) has valuation exactly $-2M$, because c is polynomial. The right side has valuation at least $-M$, because a is a nonzero constant. This is impossible. Thus no coordinate has a finite polynomial pole. A coprime rational fraction with a nonconstant denominator would have such a pole, proving the claim. □

Lemma 3.2 (Common degree across all cycles)**.** *If a periodic point exists,* $\deg c = 2m$ *for some* $m > 0$*, and every coordinate in every periodic orbit has degree* m*.*

Proof. For one orbit let m be the largest coordinate degree. If $m \leq 0$, all coordinates are constant, contrary to (3.1) and the nonconstancy of c. At a coordinate of degree $m > 0$, the right side of the recurrence has degree at most m. If $\deg c$ were smaller or larger than $2m$, the left side would have degree $2m$ or $\deg c$, respectively. Therefore $\deg c = 2m$ and the leading terms cancel. A coordinate of degree below m could not cancel the leading term of c, giving the same contradiction. The common value $m = \deg(c)/2$ depends only on the parameter, so it is the same in every orbit. This argument does not take a maximum over a periodic set not yet known to be finite. □

Proposition 3.3 (Centered parameter and global branches)**.** *If a periodic point exists,* (2.1) *holds with the stated uniqueness. With* P *fixed, every periodic coordinate is uniquely* $\sigma P + b$ *for* $\sigma \in \{1, -1\}$ *and* $b \in k$*.*

Proof. Choose one coordinate P_0 in one orbit. If y is a coordinate in any periodic orbit, subtract the recurrence at P_0 from that at y. Lemma 3.2 gives

$$\deg(y^2 - P_0^2) \leq m. \tag{3.2}$$

Both coordinates have degree m. Their leading coefficient ratio has square one, hence equals a unique $\sigma \in \{1, -1\}$. Since $2 \neq 0$, the polynomial $y + \sigma P_0$ has degree m. Factoring the left side of (3.2) shows that $y - \sigma P_0$ is constant, also allowing it to be zero. This holds across all cycles.

Apply the recurrence at P_0. Its two neighbors now show that $c = -P_0^2 + UP_0 + V$ for some $U, V \in k$. Set $P = P_0 - U/2$ and $C = V + U^2/4$. Then $c = -P^2 + C$, and the constant change in P_0 leaves every coordinate of the asserted form. Its sign and constant are unique because P is nonconstant.

If also $c = -Q^2 + D$ with nonconstant Q and constant D, then $(P - Q)(P + Q) = C - D$. If $C - D \neq 0$, both factors are nonzero constants, which would make $2P$ constant. Hence $C = D$, and the integral domain $k[t]$ gives $Q = P$ or $Q = -P$. This is a normalization of the parameter, not a change to the map. □

4 The exact signed rationality graph

Assume $c = -P^2 + C$ and fix P. Substituting $y_i = \sigma_i P + b_i$ into (3.1) and comparing the coefficients of the linearly independent polynomials P and 1 gives

$$2\sigma_i b_i = \sigma_{i+1} + a\sigma_{i-1}, \qquad b_i^2 + C = b_{i+1} + ab_{i-1}. \tag{4.1}$$

Define transition bits by $\sigma_{i+1} = \sigma_i(-1)^{\delta_i}$. The first equation is exactly the offset formula in (2.2); equivalently,

$$b_i = (a + 1)/2 - \delta_i - a\delta_{i-1}. \tag{4.2}$$

Put $\Lambda = K_a - C$. Expanding the second equation of (4.1), using $\delta_j^2 = \delta_j$, gives

$$\Lambda = a^2\delta_{i-1} + 2a\delta_i\delta_{i+1} + \delta_{i+2}. \tag{4.3}$$

For clarity, the unshifted expansion is

$$K_a - C = a^2\delta_{i-2} + 2a\delta_{i-1}\delta_i + \delta_{i+1}.$$

Thus (4.3) is an identity over $\mathbb{Z}[1/2]$ before specialization, not a characteristic-zero argument applied without checking its denominators.

Take the eight states $V = \{0,1\}^3$. A state uvw has an edge to vwz if and only if

$$\Lambda = a^2u + 2avw + z. \tag{4.4}$$

Table 2 lists all sixteen potential edges. The two outgoing labels at a state differ by one, and the two incoming labels at a prescribed successor differ by a^2. Both differences are nonzero. Thus indegree and outdegree are at most one, and the recurrent part is a disjoint union of simple directed cycles.

State	Successor for $z=0$: required Λ	Successor for $z=1$: required Λ
000	000: 0	001: 1
001	010: 0	011: 1
010	100: 0	101: 1
011	110: $2a$	111: $2a+1$
100	000: a^2	001: a^2+1
101	010: a^2	011: a^2+1
110	100: a^2	101: a^2+1
111	110: a^2+2a	111: $(a+1)^2$

Table 2: The full rationality edge table. For fixed a and $\Lambda = K_a - C$, retain exactly the edges with the displayed value of Λ. The resulting graph is a partial permutation over every field allowed by the theorem.

The bit graph has forgotten a global sign. Restore it by defining

$$E(\varepsilon; u, v, w) = \left(\varepsilon P + \frac{(-1)^v + a(-1)^u}{2}, \quad \varepsilon(-1)^v P + \frac{(-1)^w + a(-1)^v}{2}\right), \qquad \varepsilon \in \{1, -1\}. \tag{4.5}$$

Proposition 4.1 (Exact restriction and injectivity). *The sixteen labels in (4.5) are distinct points of $k[t]^2$. Every rational periodic point is among them. Their exact transition rule is*

$$H_{a,c}E(\varepsilon; u, v, w) = E(\varepsilon(-1)^v; v, w, z) \quad \Longleftrightarrow \quad \Lambda = a^2u + 2avw + z. \tag{4.6}$$

More generally, an image of a signed label can be another signed label only in the shifted form on the right of (4.6). Consequently the directed signed cycles are exactly the ordinary rational cycles of $H_{a,c}$.

Proof. The leading coefficient of the first coordinate of E determines ε relative to the fixed nonconstant P. The leading coefficient ratio of the two coordinates determines v. The first constant offset then determines u, since $a \neq 0$ and $1 \neq -1$; the second offset determines w. This proves injectivity, including the cases $a = \pm 1$ and characteristic three.

For an edge satisfying (4.4), direct substitution of (4.5), or equations (4.1)–(4.4), gives the equality in (4.6). Conversely, suppose the image is $E(\varepsilon'; u', v', w')$. The two leading coefficients first give $\varepsilon' = \varepsilon(-1)^v$ and $v' = w$. Comparing the first constant offset then gives $u' = v$, using $a \neq 0$. Set $z = w'$. The remaining constant equation is precisely (4.4). The order of this recovery matters: one constant offset alone need not determine two bits.

Proposition 3.3 and equations (4.1) show that every rational periodic point is one of these signed labels. An actual cycle therefore follows the exact signed edges. In the other direction, following a closed signed path proves the recurrence at every point and constructs an actual cycle of the original map. □

5 All-field exhaustion and ordinary periods

We now classify the bit cycles without sampling fields or determinant values. Order the eight states lexicographically and choose the least state of a cycle. The following cases use only Table 2.

Least state 000. Its self-loop has $\Lambda = 0$ and gives word 0. Otherwise the cycle starts $000 \to 001$, so $\Lambda = 1$ and the next state is 011. An edge from 011 to 111 would force $2a + 1 = 1$, contrary to $a \neq 0$. Hence it proceeds to 110, with $2a = 1$. The edge $110 \to 101$ would force $a^2 + 1 = 1$, again impossible. It must proceed to 100, with $a^2 = 1$, and then to 000. The equations $2a = 1$ and $a^2 = 1$ force $4 = 1$, hence $\operatorname{char} k = 3$ and $a = -1$; conversely these conditions satisfy all edges. The cycle is

$$000 \to 001 \to 011 \to 110 \to 100 \to 000$$

and gives word 00011.

Least state 001. If the first edge went to 011, it would have $\Lambda = 1$ and would follow the preceding forced route through 000, contradicting minimality. Therefore it goes to 010, with $\Lambda = 0$, and then to 100. The edge $100 \to 000$ would force $a = 0$. The other edge closes at 001 exactly when $a^2 + 1 = 0$. This is the word 001 and $\Lambda = 0$.

Least state 010. The edge with label zero reaches 100, whose two successors are less than 010, so it cannot belong to this cycle. The edge with label one reaches 101. Its edge to 011 would require $a^2 + 1 = 1$ and is impossible. The remaining edge returns to 010 exactly when $a^2 = 1$. This gives word 01 and $\Lambda = 1$.

Least state 011, first edge to 110. Here $\Lambda = 2a$. The edge $110 \to 100$ cannot belong to this cycle, since every successor of 100 is below 011. The edge to 101 therefore requires $a^2 + 1 = 2a$, or $(a-1)^2 = 0$. Since k is a field this means $a = 1$. The next edge closes at 011. We obtain word 011 and $\Lambda = 2$.

Least state 011, first edge to 111. Now $\Lambda = 2a + 1$. A self-loop at 111 cannot close a cycle containing 011; equality of its label with $2a+1$ would also force $a = 0$. The edge $111 \to 110$ requires $a^2 = 1$. As above, the edge $110 \to 100$ is excluded by the smaller successors of 100. Thus $110 \to 101$ requires $a^2 + 1 = 2a + 1$, hence $2a = 1$. Together these give exactly $\operatorname{char} k = 3$ and $a = -1$. The edge $101 \to 011$ closes the cycle. It gives word 0111 and $\Lambda = 2$ in characteristic three.

Remaining least states. The states 100, 101 and 110 have only smaller successors, so none can be least in a cycle. If 111 is least it must be its self-loop, with $\Lambda = (a+1)^2$ and word 1.

This exhausts every possible least state and all its allowed first edges. Replacing Λ by $K_a - C$ gives the seven parameter entries of Table 1. The equations forcing characteristic three have been retained explicitly; no characteristic-zero gcd criterion was used to discard them.

Lemma 5.1 (Sign lift and least period)**.** *Let a bit cycle have length ℓ and transition word w. If the number of ones in w is even, it lifts to two distinct ordinary cycles of least period ℓ. If the number is odd, it lifts to one ordinary cycle of least period 2ℓ. Different bit cycles have disjoint lifted point sets.*

Proof. One traversal multiplies the phase in (4.6) by $(-1)^{\sum_{j=0}^{\ell-1} w_j}$. For positive sign the two initial phases close separately. For negative sign they join after two traversals. Any earlier return of an actual point must, by injectivity of (4.5), first return to its bit state, so its time is a multiple of ℓ. It must also restore the phase. This proves leastness and shows that the two positive-sign lifts cannot be rotations of the same cycle. Distinct bit cycles have disjoint states; injectivity gives disjoint point sets. □

The words 0, 011 and 00011 have even parity. The other four have odd parity. Their lengths therefore give exactly the cycle multiplicities and ordinary periods in Table 1. Every reconstructed cycle is actual by Proposition 4.1, and every actual rational cycle has been included in the exhaustion. This proves Theorem 2.1, including the instruction to add coincident parameter rows.

6 Parameter collisions and sharp totals

The sharp bound concerns the whole periodic set, not a maximum single period. All rows that coexist must be counted.

Proof of Theorem 2.2. First take $a \notin \{1, -1\}$. Beyond fixed points and two-cycles, the only possible row is 001, which requires $a^2 = -1$. The parameters K_a and $-3K_a$ are distinct, because equality would give $(a+1)^2 = 0$. If $a^2 = -1$, the value $C = K_a$ contributes exactly two fixed points and one six-cycle, totaling eight. For other such determinants the total is at most two.

For $a = 1$, all possibly nonempty rows give

$$(C, \text{point contribution}) = (1, 2),\ (-3, 2),\ (0, 4),\ (-1, 6). \tag{6.1}$$

These constants are distinct in characteristic different from two, except for the equality $-3 = 0$ in characteristic three. That collision combines a two-cycle and a four-cycle, totaling six, and does not meet the six-point three-cycle locus. Thus the maximum for $a = 1$ is six in every allowed characteristic.

For $a = -1$, one has $K_a = 0$. At $C = 0$, two fixed points and one two-cycle give four points. The word 01 gives four points at $C = -1$. Outside characteristic three there are no further rows for this determinant. In characteristic three, 00011 contributes ten additional points at the same $C = -1$, while 0111 gives eight points at $C = 1$. The constants $0, -1, 1$ are distinct. The total is therefore fourteen exactly on (2.3).

These cases exhaust k^*. Outside characteristics two and three, the eight-point case exists exactly when -1 has a square root in k. Otherwise $a = 1$, $C = -1$ gives six points. Every asserted maximum is attained by taking $P = t$ over the stated field. Finally, every row with period greater than two requires $a = 1$, $a = -1$ or $a^2 = -1$, all implying $a^4 = 1$. □

The maximal characteristic-three locus can also be displayed without an implicit field extension. Fix any nonconstant P and write $+$ and $-$ in front of P literally as its two field signs. For $a=-1$, $C=-1$ and $\operatorname{char} k=3$, the coordinate words in Table 3 are periodic under the recurrence. Consecutive entries, cyclically paired, are the ordinary points. The same P is used in all three rows.

Least period	Cyclic coordinate word
4	$(P+1,\ P-1,\ -P+1,\ -P-1)$
5	$(P+1,\ P,\ P,\ P-1,\ -P)$
5	$(-P+1,\ -P,\ -P,\ -P-1,\ P)$

Table 3: All fourteen points on the equality locus: a word $(y_0,\ldots,y_{\ell-1})$ represents the ℓ ordered pairs (y_i,y_{i+1}), with indices modulo ℓ. Repeated individual coordinates do not merge ordered pairs. The words are obtained from (2.2) using 01 and 00011; Proposition 4.1 proves their mutual disjointness.

Corollary 6.1 (Ordinary return counts)**.** *Let m_j be the number of cycles of length j obtained by adding the applicable rows in Table 1. Then*

$$\#\operatorname{Fix}(H_{a,c}^n)(k(t))=\sum_{j\mid n} jm_j,\qquad \zeta_{H,k(t)}(z)=\prod_j(1-z^j)^{-m_j}. \tag{6.2}$$

The product is finite; when there are no periodic points it equals one.

Proof. A cycle of least period j contributes all its j points at time n exactly when $j\mid n$. This proves the first identity. For the ordinary Artin–Mazur definition, as formal power series at zero,

$$\log\zeta_{H,k(t)}(z)=\sum_{n\geq 1}\frac{z^n}{n}\sum_{j\mid n} jm_j=-\sum_j m_j\log(1-z^j).$$

Exponentiation gives the second identity. ☐

For example, the equality locus has $\zeta_{H,k(t)}(z)=(1-z^4)^{-1}(1-z^5)^{-2}$. This records the same finite rational periodic set and is not a second classification problem or a Frobenius zeta of a variety.

7 Scope and reproducibility

The proof has reduced a global rationality question to a finite partial permutation, while preserving the constant field and ordinary time. Its essential steps are finite-pole integrality, degree rigidity across all cycles, exact offset equations, injective labels and an all-field least-state exhaustion. In particular, neither an unrestricted local shift nor a finite prime sample supplies the seven-row theorem.

The accompanying exact-arithmetic script `certify_symbol_graph.py` is supplementary corroboration. It enumerates the nineteen simple cycles in the unrestricted eight-state shift graph, derives their integer-polynomial edge labels, checks the sixteen reconstruction identities over $\mathbb{Q}[a]$, and verifies fourteen distinct points over $\mathbb{F}_3[t]$. The source package records its actual execution and an independent internal review of the full proof. The script is not used to prove the finite-pole argument, the across-cycle degree reduction, or the classification over arbitrary k; all these arguments are typeset above. No manuscript build is treated as an additional mathematical verification.

The hypotheses delimit the conclusion. Nonconstant determinants would enter both the finite valuations and the degree comparison. Rational parameters with additional poles would require a different global argument. For constant c the positive-degree reduction fails, and in characteristic two the two signs and division by two are unavailable. We make no classification claim for those families, for arbitrary finite extensions of $k(t)$, or for the number-field uniform boundedness conjecture.

Finally, the finite-cycle zeta in (6.2) is a source counting identity. It gives no target Euler factors, root numbers, automorphy, zero/divisor correspondence or Hilbert–Pólya realization. Those questions require additional arithmetic structure not supplied by this rational periodic atlas.

References

[1] Kenneth Allen, David DeMark, and Clayton Petsche. Non-Archimedean Hénon maps, attractors, and horseshoes. arXiv:1610.04271v3, 2018. URL https://arxiv.org/abs/1610.04271v3. Version 3, 6 February 2018; initially submitted 13 October 2016.

[2] Patrick Ingram. Canonical heights for Hénon maps. *Proceedings of the London Mathematical Society*, 108(3):780–808, 2014. doi: 10.1112/plms/pdt026. Text comparison uses arXiv:1111.3609v1.